

Grad Store Builder

Version One | Advanced Implementation Plan

Technical Architecture Deck | 7 Gates | 2026 Stack

An authenticated, multi-tenant AI e-commerce store builder
built on Next.js 16 • Vercel Workflows • AI SDK v6 • Supabase RLS

`docs/latex/01-version-one-technical-deck.tex` • 2026

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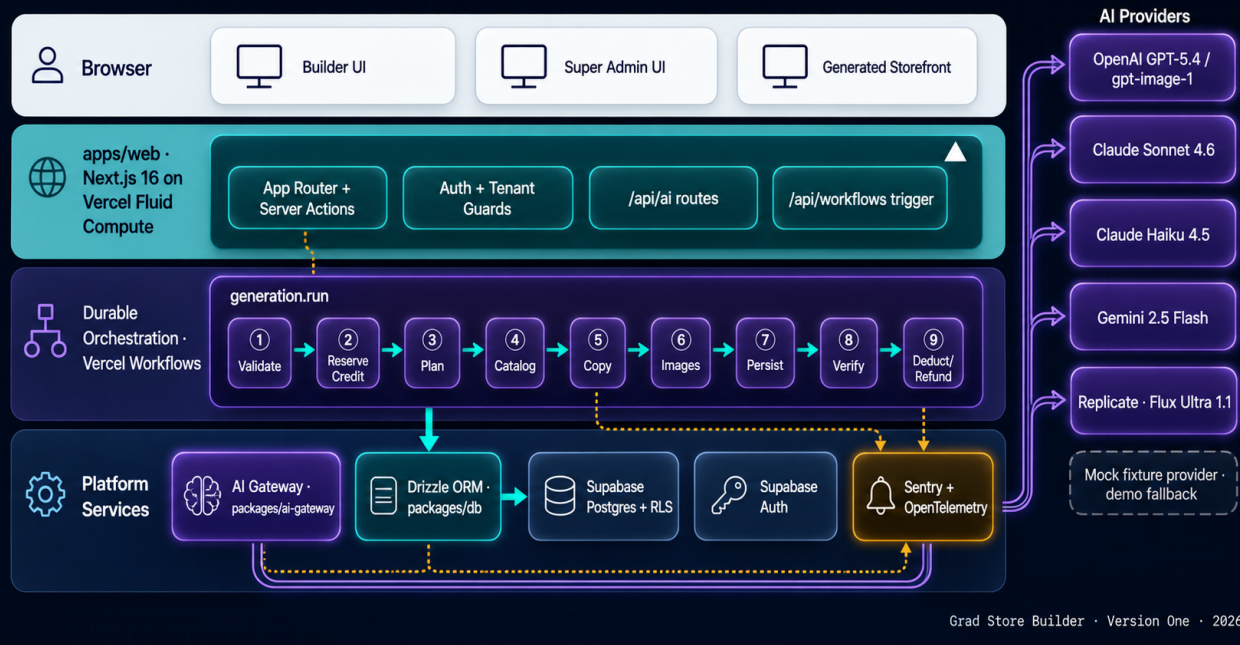
Reading order. The deck mirrors the original image-prompt order in docs/prompts/09-version-one-gate-diagram-image-prompts.md. Each section is one landscape page: image on the left, technical narrative on the right.

System Architecture

Browser • Next.js 16 on Vercel • Vercel Workflows • AI Gateway • Supabase + RLS • Sentry

End-to-end view
Master diagram

System Architecture — Grad Store Builder V1



Stack at a glance

- **App** Next.js 16 App Router on Vercel Fluid Compute
- **Auth** Supabase Auth (@supabase/ssr) cookies-based SSR
- **DB** Supabase Postgres + RLS via Drizzle pgPolicy
- **Orchestration** Vercel Workflows (GA April 2026)
- **AI** AI SDK v6 • `Output.object()` • ToolLoopAgent
- **Models** GPT-5.4 • Sonnet 4.6 • Haiku 4.5 • Gemini 2.5 Flash
- **Images** gpt-image-1 + Flux Ultra 1.1 (Replicate)
- **Observability** Sentry + @sentry/opentelemetry
- **Verify** Playwright + Chrome DevTools MCP

Why this shape

- One Next.js app today, monorepo packages already split for future apps
- RLS at the DB *enforces* multi-tenancy — no client trust
- Workflows make the multi-step AI pipeline durable and idempotent
- Mock provider on every AI task = zero-network demo safety

Gate 0 • Source Confirmation

Pin every library version. Write nothing else.

1 day

v0.0-gate-0-source-confirmed

Gate 0 — Source Confirmation • 1 day

Pin every version. Write nothing else.



What happens

- Run **Context7 MCP** against every key library
- Run **OpenAI Docs MCP** for structured-output + image-1
- Pin exact versions in the planned package.json
- Draft ADR 0003 (stack lock-in) and ADR 0004 (AI routing)

Libraries in scope

- next.js@16, @supabase/ssr, drizzle-orm
- ai@6, @ai-sdk/openai | anthropic | google
- @vercel/workflows, shadcn-ui (cli v4)
- @sentry/nextjs, playwright

Done when

Pinned versions recorded • confirmation log written at docs/logs/2026-05-XX-source-confirmation.md • ADRs 0003 + 0004 drafted.

Gate A • Foundation

Empty app deploys to Vercel preview. Login works. RLS proven.

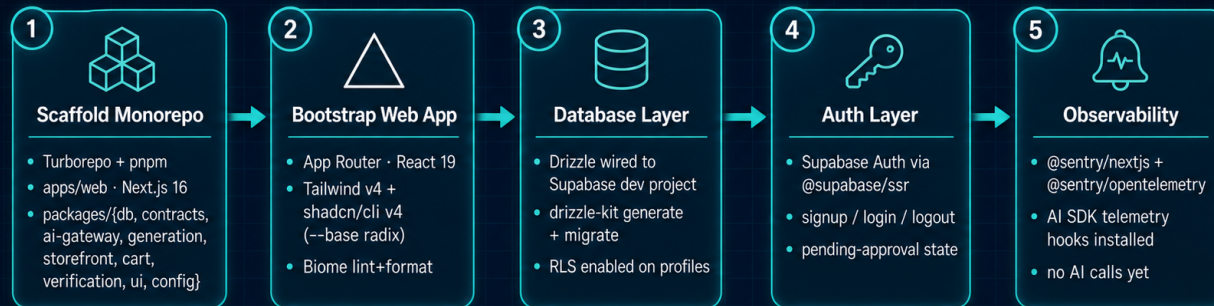
3–4 days

v0.1-gate-a-foundation-pass

Gate A — Foundation • 3–4 days

Empty app deploys to Vercel preview • login works • RLS proven.

Branch: `gate-a/foundation-scaffold`



Done when

- ☒ empty app deploys to Vercel preview
- ☒ login works
- ☒ Sentry receives a test error
- ☒ negative cross-user test on profiles passes

Tag on pass: `v0.1-gate-a-foundation-pass`

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The five stages

1. **Scaffold** Turborepo + pnpm • all 9 packages created empty
2. **Bootstrap** Next.js 16 • Tailwind v4 • shadcn cli v4 • Biome
3. **DB** Drizzle wired to a fresh Supabase *dev* project
4. **Auth** Supabase Auth via `@supabase/ssr` • pending-approval state
5. **Observability** Sentry + OTel + AI SDK telemetry hooks installed (no AI calls yet)

Done when

Empty app deploys to Vercel preview • login works • Sentry receives a test error • negative cross-user test on profiles passes.

Risk & mitigation

Forgetting RLS on profiles. Mitigation: the cross-user negative test is the gate's fail-stop — if it doesn't pass, Gate A doesn't tag.

Gate B • Tenancy • Credits • Admin

RLS at the DB. Immutable ledger. Lazy Super Admin queue.

4–5 days

v0.2-gate-b-tenancy-credits-pass

Gate B — Tenancy • Credits • Admin • 4–5 days

RLS at the DB • immutable ledger • lazy Super Admin.

Branch: gate-b/credits-ledger

Tenancy Schema (Drizzle + pgPolicy)

profiles	id	uuid	PK
	email	text	UNIQUE
	created_at	timestampz	DEFAULT now()

organizations	id	uuid	PK
	name	text	NOT NULL
	created_at	timestampz	DEFAULT now()

organization_members	id	uuid	PK
	organization_id	uuid	NOT NULL
	user_id	uuid	NOT NULL
	role	text	NOT NULL
	created_at	timestampz	DEFAULT now()

admin_actions	id	uuid	PK
	admin_id	uuid	NOT NULL
	action	text	NOT NULL
	target_type	text	NOT NULL
	target_id	uuid	NOT NULL
	metadata	jsonb	NOT NULL
	created_at	timestampz	DEFAULT now()

memberships	id	uuid	PK
	user_id	uuid	NOT NULL
	organization_id	uuid	NOT NULL
	status	text	NOT NULL
	created_at	timestampz	DEFAULT now()

RLS policy: organization_id IN (SELECT organization_id FROM organization_members WHERE user_id = (SELECT auth.uid()))

Credits System (Immutable Ledger)

credit_balances	organization_id	uuid	PK
	balance	numeric	NOT NULL
	updated_at	timestampz	NOT NULL

credit_ledger	id	uuid	PK
	organization_id	uuid	NOT NULL
	entry_type	text	NOT NULL
	amount	numeric	NOT NULL
	reference_id	uuid	NOT NULL
	metadata	jsonb	NOT NULL
	created_at	timestampz	DEFAULT now()

$\sum \text{ledger.amount per org} == \text{credit_balances.balance}$

append-only • immutable

← reserveCredits()
← deductCredits()
← refundCredits()
← grantCredits()

Lazy Super Admin UI

server-side super-admin guard • never trust client
Pending Users Queue
Credit Grant Form
Admin Action Audit Log

Done when

- ☒ pending user blocked
- ☒ approved user lands in builder
- ☒ ledger sum equals balance for every org
- ☒ cross-tenant access tests pass

Tag on pass: v0.2-gate-b-tenancy-credits-pass

Three pillars

- Tenancy schema:** profiles, organizations, organization_members, admin_actions, memberships — every table guarded by Drizzle pgPolicy
- Credits ledger:** credit_balances (materialized) + credit_ledger (append-only); helps reserveCredits / deductCredits / refundCredits / grantCredits
- Lazy Super Admin:** pending-user queue, credit-grant form, audit log — minimal pages, server-side guard checked on every privileged action

Invariants

- $\sum \text{ledger.amount per org} == \text{credit_balances.balance}$
- No row reachable across orgs (cross-tenant probes fail)

Done when

- Pending user blocked • approved user lands in builder
- ledger sum equals balance for every org • cross-tenant tests pass.

Gate C • AI Gateway + Brand Intake

AI SDK v6 •

• per-task routing • mock fallback

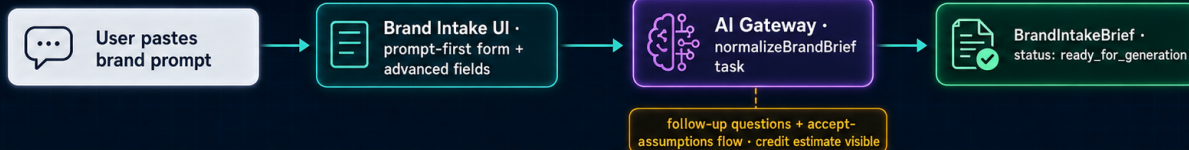
5–6 days

v0.3-gate-c-ai-intake-pass

Gate C — AI Gateway + Brand Intake • 5–6 days

AI SDK v6 • Output.object() • per-task routing • mock fallback.

Brand Intake Flow



AI Gateway Internals

packages/ai-gateway



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Brand intake flow

prompt → Brand Intake UI →
 AI Gateway.normalizeBrandBrief
 → BrandIntakeBrief (status:
 ready_for_generation) in <30 s.

Per-task routing matrix (excerpt)

- normalizeBrandBrief → **Sonnet 4.6** --> GPT-5.4
- createStorePlan → **GPT-5.4** --> Sonnet 4.6
- generateCatalog → **GPT-5.4** --> Gemini 2.5 Flash
- generateCopyBlocks → **Sonnet 4.6** --> GPT-5.4
- reviewSafety → **Haiku 4.5** --> GPT-5-nano
- generateProductImage → **gpt-image-1 + Flux Ultra** --> static placeholder
- **All tasks have a mock-provider fallback.**

Done when

Sample brief returns ready_for_generation in <30 s •
 prompt-injection attempts produce inert data • mock
 provider drives full demo flow with zero network.

Gate D • Store Generation Pipeline

Vercel Workflow

• 9 retryable steps • credits deducted only on success

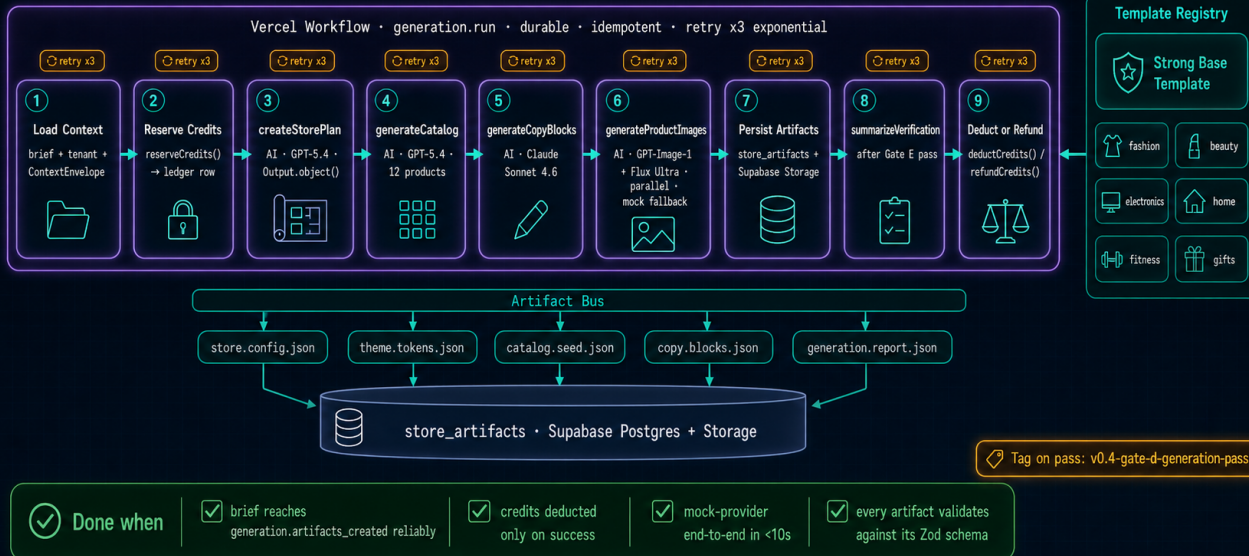
6–7 days

v0.4-gate-d-generation-pass

Gate D — Store Generation Pipeline • 6–7 days

Vercel Workflow • generation.run • 9 retryable steps • credits deducted only on success.

Branch: gate-d/workflows-generation-run



The nine steps

1. Load Context (brief + tenant + ContextEnvelope)
2. Reserve Credits (→ ledger row)
3. **createStorePlan** (GPT-5.4 • Output.object())
4. **generateCatalog** (12 products)
5. **generateCopyBlocks** (Sonnet 4.6)
6. **generateProductImages** (parallel • mock fallback)
7. Persist Artifacts (→ store_artifacts + Storage)
8. **summarizeVerification** (after Gate E pass)
9. Deduct or Refund (terminal status)

Artifact bus

store.config.json • theme.tokens.json •
 catalog.seed.json • copy.blocks.json •
 generation.report.json

Template registry: 1 strong base + 6 presets (fashion, beauty, electronics, home, fitness, gifts).

Done when

Brief reaches generation.artifacts_created reliably

- credits deducted only on success
- mock-provider end-to-end in <10 s
- every artifact validates against its Zod schema.

Gate E • Storefront + Cart + Mock Checkout + Verification

Pure renderer over artifacts • platform-owned cart • 9 verification checks block

5–6 days

v0.5-gate-e-storefront-verify-pass

Gate E — Storefront • Cart • Mock Checkout • Verification • 5–6 days

Branch: gate-e/verification-runner

Pure renderer over artifacts • platform-owned cart • 9 verification checks block demo_ready.

Storefront Render Path



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Render path

store_artifacts → packages/storefront (pure renderer) → Home, Listing, Detail, Cart, Mock Checkout, Confirmation. **No AI in the render loop.**

Verification runner — 9 checks

1. Schema presence
2. Tenant ownership probe (impersonate other org)
3. Credit-ledger invariant
4. All required routes return 200
5. Cart add/remove/qty/subtotal (Playwright)
6. Mock checkout success + cancel
7. Responsive @ 360 / 768 / 1280 / 1920
8. Playwright a11y axe scan
9. Prompt-injection / secret-leak grep on artifacts

Done when

100% of verification checks pass • failed verification keeps store out of demo_ready (credits refunded).

Gate F • Demo Readiness + Super Admin Polish

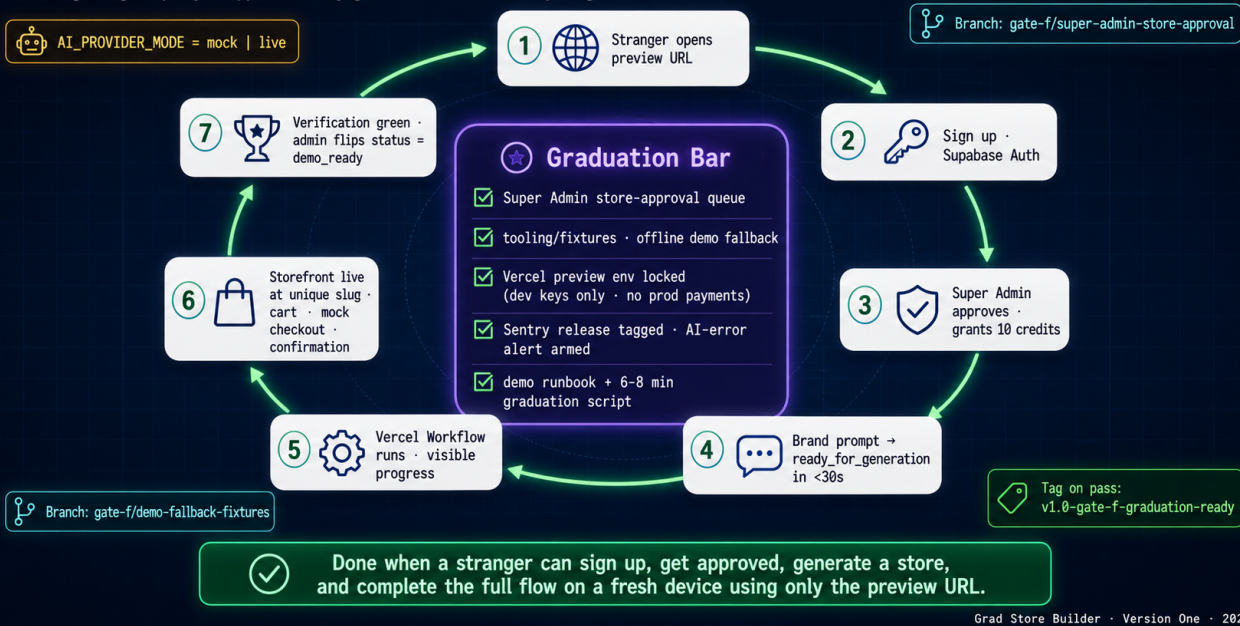
A stranger signs up • you approve • they generate a store • everything works on a fresh device

3–4 days

v1.0-gate-f-graduation-ready

Gate F — Demo Readiness • Graduation Bar • 3–4 days

A stranger signs up • you approve • they generate a store • everything works on a fresh device.



Demo-day journey (clockwise)

1. Stranger opens preview URL
2. Sign up via Supabase Auth
3. Super Admin approves & grants 10 credits
4. Brand prompt → `ready_for_generation` in <30 s
5. Vercel Workflow runs with visible progress
6. Storefront live at unique slug • cart • mock checkout • confirmation
7. Verification green • admin flips status to `demo_ready`

Graduation bar checklist

- Super Admin store-approval queue
- tooling/fixtures/ offline demo fallback
- Vercel preview env locked (dev keys, no prod payments)
- Sentry release tagged • AI-error alert armed
- Demo runbook + 6–8 min graduation script

Done when

Stranger completes the full flow on a fresh device using only the preview URL • `AI_PROVIDER_MODE = mock` | `live` switch works.

Bonus • AI Task Routing Matrix

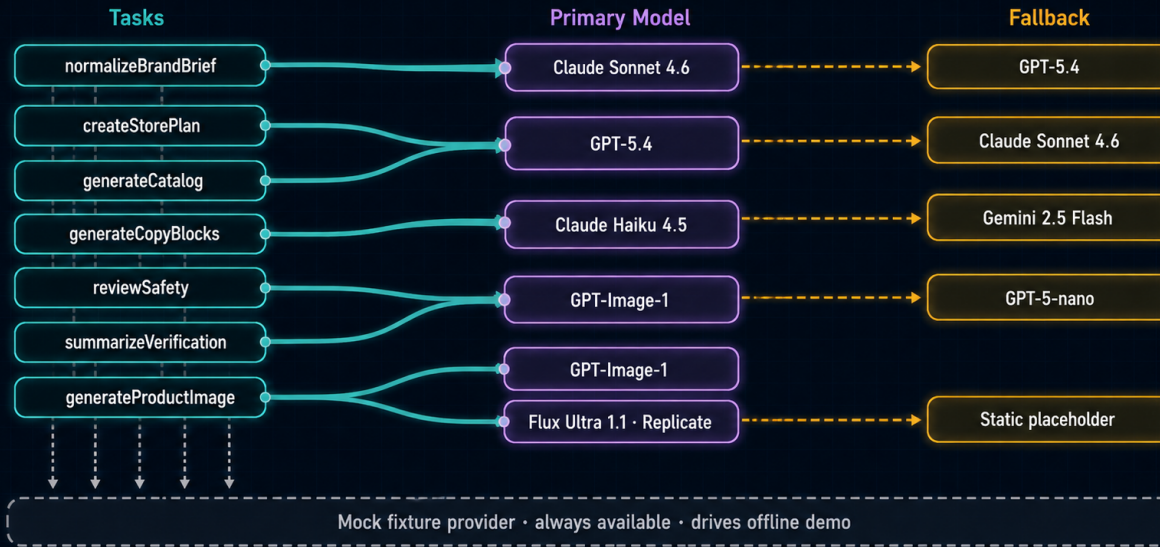
Refs: docs/decisions/0004-ai-task-routing-matrix.md

Per-task primary model + documented fallback + always-on mock

Reference

AI Task Routing Matrix • 2026

Per-task primary model • documented fallback • always-available mock provider.



Refs: docs/decisions/0004-ai-task-routing-matrix.md

Routing rules summary

- Reasoning-heavy structured tasks → **Claude Sonnet 4.6**
- Long-instruction JSON generation → **GPT-5.4**
- Cheap utility classification → **Claude Haiku 4.5**
- Photoreal images → **Flux Ultra 1.1** on Replicate
- Composition/text-on-image → **gpt-image-1**
- Catalog scale-out fallback → **Gemini 2.5 Flash** (cheapest)
- Mock provider** drives offline demo for every task

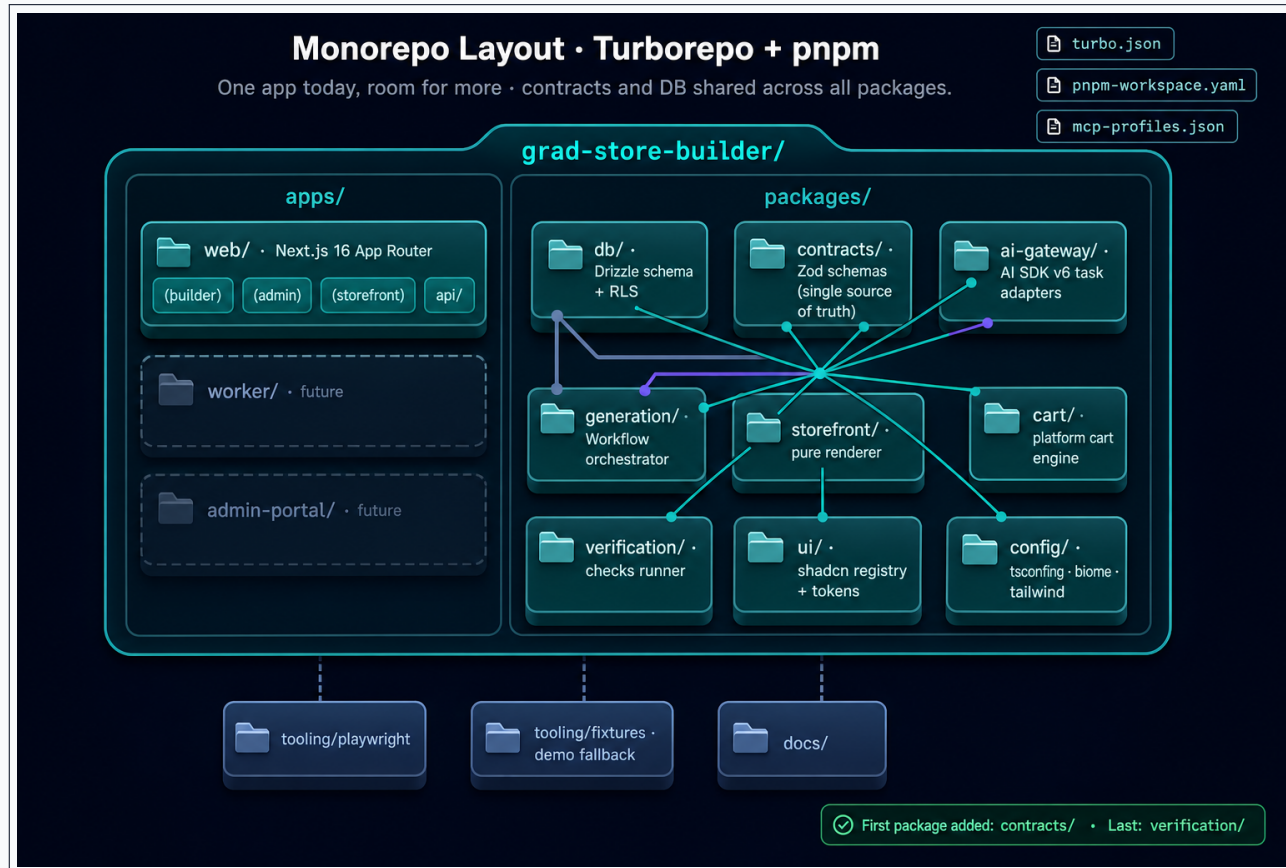
Why two image models

gpt-image-1 composes layouts and renders text on labels reliably; Flux Ultra 1.1 wins on photoreal hero shots. Routing picks based on the artifact `image.kind`: hero, lifestyle, packshot, label.

Bonus • Monorepo Layout

One app today, room for more • + shared across all packages

Turborepo + pnpm
Reference



Today

- **apps/web** — Next.js 16 App Router
- **packages/contracts** — Zod schemas (single source of truth, depended on by every package)
- **packages/db** — Drizzle schema + RLS policies + migrations
- **packages/ai-gateway** — AI SDK v6 task adapters
- **packages/generation** — Workflow orchestrator
- **packages/storefront** — pure renderer
- **packages/cart** — platform cart engine
- **packages/verification** — checks runner
- **packages/ui** — shadcn registry + tokens
- **packages/config** — tsconfig • biome • tailwind

Future (no rewrites)

apps/worker • apps/admin-portal • apps/mobile all slot in beside apps/web, sharing the same packages.

Appendix • Gate Roadmap at a Glance

Gate	Theme	Days	Branch	Tag on pass
0	Source Confirmation	1	gate-0/source-confirmation	v0.0-gate-0-source-confirmed
A	Foundation	3–4	gate-a/foundation-scaffold	v0.1-gate-a-foundation-pass
B	Tenancy + Credits + Admin	4–5	gate-b/credits-ledger	v0.2-gate-b-tenancy-credits-pass
C	AI Gateway + Brand Intake	5–6	gate-c/ai-gateway-tasks	v0.3-gate-c-ai-intake-pass
D	Store Generation Pipeline	6–7	gate-d/workflows-generation-run	v0.4-gate-d-generation-pass
E	Storefront + Verification	5–6	gate-e/verification-runner	v0.5-gate-e-storefront-verify-pass
F	Demo Readiness	3–4	gate-f/super-admin-store-approval	v1.0-gate-f-graduation-ready
Total working days to graduation bar		≈ 33		

Done means. A stranger signs up at the preview URL. Super Admin approves them and grants 10 credits. They submit a brand prompt; AI returns a structured brief in <30 s. The Vercel Workflow runs the 9 generation steps with visible progress. A storefront with home, listing, detail, cart, mock checkout, and confirmation is live at a unique slug. The verification report is green; admin store-approval flips status to `demo_ready`. Sentry shows traces, token costs, and zero unresolved errors. The same demo runs offline using fixtures.